REVIEW AND PRACTICE PRACTICAL

1. **Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.**
   - ➢

   Configuration Steps:
   1. Opened Cisco Packet Tracer and created a new project file.
   2. Added one Wireless Router from:
      - Network Devices → Wireless Devices → Wireless Router
   3. Added three PCs from:
      - End Devices → PC
   4. Renamed the devices as:
      - PC1
      - PC2
      - PC3
   5. Connected the three PCs to the wireless router using wireless connections/copper straight-through cables.
   6. Configured the wireless router:
      - Opened Router → Config tab.
      - Set the SSID name for WiFi network.
      - Enabled wireless service.
   7. Configured each PC:
      - Opened PC → Desktop → IP Configuration.
      - Selected DHCP to automatically assign IP addresses.
   8. Verified the network connection by checking that all devices were connected successfully.

2. **Assign static Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.**
   - ➢

   Configuration Steps:
   1. Opened Cisco Packet Tracer and created a new project.
   2. Added one wireless router and three PCs.
   3. Connected PCs with the wireless router.
   4. Configured router SSID as HomeWiFi.
   5. Connected PCs to the WiFi network.
   6. Assigned IP addresses using DHCP.

7. Verified connectivity using ping command.
8. Saved the project as HomeWiFiNetwork.pkt.

3. **IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.**
   - **Configuration Steps:**
     1. Opened Cisco Packet Tracer and created a new project file.
     2. Added one Wireless Router and three PCs from the device list.
     3. Connected all three PCs to the wireless router using appropriate connections.
     4. Configured the router with a static IP address:
        - IP Address: 192.168.10.1
        - Subnet Mask: 255.255.255.0
     5. Configured PC1 with a static IP address:
        - IP Address: 192.168.10.2
        - Subnet Mask: 255.255.255.0
        - Default Gateway: 192.168.10.1
     6. Configured PC2 with a static IP address:
        - IP Address: 192.168.10.3
        - Subnet Mask: 255.255.255.0
        - Default Gateway: 192.168.10.1
     7. Configured PC3 with a static IP address:
        - IP Address: 192.168.10.4
        - Subnet Mask: 255.255.255.0
        - Default Gateway: 192.168.10.1
     8. Verified the IP configuration on each PC using the ipconfig command.
     9. Tested connectivity between PCs using the ping command to ensure successful communication.

4. **Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.**
   - **Configuration Steps:**

1. Opened the HomeWiFiNetwork.pkt file in Cisco Packet Tracer.
2. Verified that all devices were connected properly:
    - PC1
    - PC2
    - PC3
    - Wireless Router
3. Checked the IP addresses of all devices:

Device	IP Address
Router	192.168.10.1
PC1	192.168.10.2
PC2	192.168.10.3
PC3	192.168.10.4

4. Opened PC1:
    - Clicked on PC1.
    - Selected Desktop → Command Prompt.
    - Entered the ping command:

    ping 192.168.10.3

- Verified that PC1 successfully communicated with PC2.

5. Tested PC1 connection with PC3:

    ping 192.168.10.4

6. Tested PC1 connection with the router:

    ping 192.168.10.1

7. Opened PC2 Command Prompt and tested communication with PC3:

    ping 192.168.10.4

8. Checked the ping results. Successful communication displayed messages like:

    Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
    Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

9. Confirmed that all PCs and the router were able to communicate successfully.

5. Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

➢

Step 1: Create a New Packet Tracer Project

1. Open **Cisco Packet Tracer**.
2. Select **File → New** to create a new project.
3. Save the project file with the name:
   HomeWiFiNetwork.pkt

Step 2: Add Network Devices

1. Added one **Wireless Router** from:
   Network Devices → Wireless Devices → Wireless Router
2. Added three PCs from:
   End Devices → PC
3. Renamed the PCs as:

- PC1
- PC2
- PC3

Step 3: Connect Devices

1. Connected all PCs with the wireless router using appropriate connections.
2. Verified that all device connections were active.

Screenshot 1:
(Paste final Packet Tracer topology screenshot here)

Step 4: Configure Router IP Address

1. Opened the Wireless Router.
2. Selected the **Config → LAN** option.
3. Assigned the router IP address:

IP Address: 192.168.10.1
Subnet Mask: 255.255.255.0
The router IP was used as the default gateway for all PCs.

Step 5: Configure PC IP Addresses

PC1 Configuration:
Opened:
PC1 → Desktop → IP Configuration → Static
Entered:
IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Screenshot 2:
(Paste PC1 IP Configuration screenshot here)

PC2 Configuration:
Entered:
IP Address: 192.168.10.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1
Screenshot 3:
(Paste PC2 IP Configuration screenshot here)

PC3 Configuration:
Entered:
IP Address: 192.168.10.4
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1
Screenshot 4:
(Paste PC3 IP Configuration screenshot here)

Step 6: Test Network Connectivity

1. Opened PC Command Prompt:
   Desktop → Command Prompt
2. Tested communication between devices using ping commands.
   Example:
   ping 192.168.10.3
   ping 192.168.10.4
   ping 192.168.10.1
3. Received successful reply messages from other devices.
   Screenshot 5:
   (Paste successful ping result screenshot here)

IP Address Table:

Device	**IP Address**		**Subnet Mask**	**Default Gateway**
Router	192.168.10.1		255.255.255.0	-
PC1	192.168.10.2		255.255.255.0	192.168.10.1

PC2	192.168.10.3		255.255.255.0	192.168.10.1
PC3	192.168.10.4		255.255.255.0	192.168.10.1